

Suppose the regression model

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + \beta_3 x_i z_i + \epsilon_i.$$

1. Find the marginal effect of x .
2. Show that if the coefficient estimates are unbiased, then the plug-in marginal effect estimate is unbiased. (Would this generalize to, say $\frac{\beta_1}{\beta_3}$?)
3. Find the variance of the plug-in marginal effect estimate.